Has Large-Load Growth from AI Data Centers Made Texas Electricity Harder to Predict?

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(a) Problem and Questions of Interest

Electricity demand forecasting in Texas has historically rested on a weather-driven relationship.

Temperature, time of day and day of week explain most variation in system load.

- **ERCOT (Electric Reliability Council of Texas)** is now processing an unprecedented volume of large-load interconnection requests, reported at more than 233 GW as of Dec-2025 and up roughly 300% over the prior year end, with more than 70% **attributed to data centers**. This demand runs continuously and is largely insensitive to weather.

https://www.utilitydive.com/news/ercots-large-load-queue-jumped-almost-300-last-year-official/808820/

Preliminary analysis:

- Hourly ERCOT operating data from 2019 to 2026 indicates this shift is already observable in the load shape. Average daily minimum demand rose 41.2% while average daily peak demand rose 33.9%. The ratio of daily minimum to daily maximum load increased from 0.696 to 0.728, rising monotonically from 2022 onward. The load floor is rising faster than the ceiling, which is consistent with growing always-on demand. 2026 covers January to September only, and shoulder months typically show flatter daily profiles, so the full-year figure is likely equal or higher.
- The growth is geographically concentrated. Subregional demand data shows the Far West zone rising from 7.68% to 13.46% of ERCOT demand between 2019 and 2025, and the North zone from 2.24% to 3.44%, while the Coast and North Central zones fell in share despite absolute growth. These are the regions where large-load interconnection has concentrated.

This analysis asks:

1. How accurately can ERCOT demand and net load be predicted 24 hours ahead using only public weather, calendar and lagged-load features?
2. Can grid stress hours, defined as net load in the upper tail of its seasonal distribution, be classified 24 hours in advance?
3. Has the predictability of Texas load from public data changed as load composition has shifted? Demand and net load are examined separately, since demand reflects consumption behaviour while net load additionally absorbs the variability of a wind and solar fleet that has grown several-fold.

Prior Work:

1. Skiles, Rhodes and Webber (2023) characterise ERCOT seasonal peak evolution through 2021 using linear regression on annual peaks, predating the large-load wave and examining peaks rather than hourly load shape. https://arxiv.org/abs/2302.01304
2. Optimisation studies simulate data center integration through unit commitment and flexibility models, but treat data center load characteristics as exogenous assumptions, in at least one case assuming a 90% load factor drawn from secondary sources. https://arxiv.org/abs/2509.07218
3. A third analyses interconnection queues and long-term projections, where ERCOT has itself cautioned that much projected demand may not materialise on schedule and independent analysts have questioned forecast inflation (Goggin and Gramlich, 2025). https://gridstrategiesllc.com/wp-content/uploads/Grid-Strategies-National-Load-Growth-Report-2025.pdf

No identified study applies supervised learning to the observed public operating record to test whether this compositional change has altered the predictability of Texas load. This project addresses that gap.

(b) Data Available

1. ***EIA-930 Hourly Electric Grid Monitor.*** Form EIA-930 collects hourly demand, day-ahead demand forecast, net generation, net generation by energy source and net interchange from the 65 balancing authorities operating the Lower 48 grid, published as six-month bulk CSV files requiring no authentication.
    a. https://www.eia.gov/electricity/gridmonitor/
2. ***EIA-930 subregional demand.*** SUBREGION files disaggregate ERCOT demand into constituent weather zones from July 2018 onward, supporting geographically resolved features.
    a. https://www.eia.gov/electricity/gridmonitor/sixMonthFiles/EIA930_SUBREGION_2026_Jan_Jun.csv
3. ***Open-Meteo historical weather archive***. ERA5 reanalysis provides gap-free hourly temperature, dew point, wind speed, cloud cover and shortwave radiation from 1940 to present at arbitrary coordinates, without authentication. Eight coordinates corresponding to the ERCOT weather zone stations identified by Skiles et al. (2023) will be demand-weighted into system aggregates, extending their method using subregional demand data
    a. https://open-meteo.com/en/docs/historical-weather-api

Data acquired and validated: Sixteen half-year files filtered to ERCO for 1 January 2019 to 18 September 2026 give **67,631 hourly observations**, 99.998% of possible hours, with 48 missing. Net load averages 34,127 MW across a 6,978 to 75,208 MW range. One integrity issue was found and resolved: EIA renamed fuel-source columns in late 2024, splitting wind and solar into with-battery and without-battery variants, so exact-name matching silently nulls all prior records.

After feature engineering, the modelling table will contain approximately 67,600 rows and 60 derived columns, giving n x p of roughly 4.1 million.

(c) Possible Approaches

- **Feature engineering:** Calendar features with cyclical encodings and holiday flags. Demand-weighted temperature, plus cooling and heating degree hours. Nonlinear temperature terms and temperature-by-hour interactions.
- **Regression models:** Ordinary least squares as baseline, then **ridge** and **lasso** with penalties tuned by cross-validation. Regularization is needed because lagged and weather features are heavily collinear.
- **Classification models:** Regularized **logistic regression**, **support vector machines** with an RBF kernel, **random forests** and **gradient boosting**. Tree ensembles are included because load response to temperature is nonlinear and varies by hour.
- **Deep learning:** An **LSTM** on 168-hour sequences, to test whether recurrent structure beats engineered lags.
- **Unsupervised:** k-means on normalised daily load shapes, with PCA for visualisation. This gives independent evidence of structural change.
- **Validation:** Blocked rolling-origin cross-validation, not random k-fold, since random splits train on future observations. Both are reported to quantify the resulting optimism.

- **Metrics:** Regression assessed by MAE, RMSE and MAPE, benchmarked against ERCOT's published forecast on identical hours. Classification assessed by confusion matrix, precision, recall, F1 and AUC-PR, since positives are expected in the low single digits depending on threshold.

(d) Key Uncertainties

1. **Attribution is limited:** ERCOT does not publish metered load by customer class, so large-load growth cannot be proven to cause the observed flattening. Findings are framed as consistent with, not proof of, that mechanism. The same analysis will be run on other balancing authorities as a comparison.
2. **Direction is uncertain:** Always-on load removes weather signals but is itself highly regular. Demand and net load predictability may diverge; whichever way the result falls, it will be reported.
3. **Confounders:** Population growth, heating electrification (Skiles et al., 2023), winterisation reforms, crypto load and solar and battery buildout all move load over this window. Generation mix variables are available and will be controlled for.
4. **Threshold choice:** results reported at three stress thresholds.
5. **Benchmark caveat:** ERCOT has private scheduling data, so the comparison measures the public-data gap rather than a fair contest.
6. **The window may be too short:** Large-load growth accelerated only recently, and five to seven years of operating data may not be sufficient to detect a predictability effect even if one is developing.

(e) References

1. Busby, J.W., Baker, K., Bazilian, M.D., Gilbert, A.Q., Grubert, E., Rai, V., Rhodes, J.D., Shidore, S., Smith, C.A. and Webber, M.E. (2021). Cascading risks: Understanding the 2021 winter blackout in Texas. Energy Research & Social Science 77, 102106. https://doi.org/10.1016/j.erss.2021.102106
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Generative AI (Claude, Anthropic) assisted in scoping the research question, identifying public data sources and drafting this document. All data acquisition, validation and preliminary analysis were executed and verified by the author.